

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

Third Semester, September, 2025

Paper Code: ECC-207**Subject: Digital Logic and Computer Design****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.**

Q. No.	Question	Max. Marks	CO(s)
1. (a)	Convert: 1. $[3643]_8 = [\quad]_{16}$ 2. Decimal 5678 to Excess-3 code.	2	CO1
(b)	Reduce the Boolean expression: $AB + (AC)' + AB'C(AB + C)$. [Note: $(AC)'$ denotes complement of AC; B' denotes complement of B; prime notation denotes logical negation].	2	CO1
(c)	Compare Asynchronous and Synchronous counters.	2	CO2
(d)	What is meant by race around condition in flip flops?	2	CO2
(e)	Draw Serial In Serial Out shift register.	1	CO2
(f)	Subtract $(1010)_2$ from $(1111)_2$ using 2's complement.	1	CO1
2. (a)	Explain and design a full subtracter.	5	CO1
(b)	Design a 3-bit Ripple Down counter and draw its waveform.	5	CO2
3. (a)	Implement the following Boolean function using 8:1 MUX: $F(A, B, C, D) = \sum m(1, 2, 5, 6, 7, 8, 10, 12, 13, 15)$	5	CO1
(b)	Realise S-R flip flop using J-K flip flop.	5	CO2
4. (a)	Describe the design and working of Master-Slave J-K flip flop.	5	CO2
(b)	Draw and explain 3:8 decoder.	5	CO1

****** End of Question Paper ******